

From Ledgers to Ledgers

Reimagining ESG Auditing in the Age of AI and Blockchain-Driven Accountability

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ABSTRACT

This study presents a transformative vision for ESG auditing by proposing a smart, technology-enabled model that leverages Artificial Intelligence (AI) and Blockchain to enhance the credibility, transparency, and efficiency of non-financial assurance. As Environmental, Social, and Governance (ESG) reporting becomes a regulatory and reputational imperative worldwide, traditional manual assurance methods struggle to meet the demand for real-time validation and data integrity. This paper addresses that gap by introducing a five-layer conceptual framework for smart ESG auditing.

The model begins with automated data acquisition through IoT sensors and enterprise systems, followed by blockchain-based data validation to guarantee immutability and traceability. It then employs AI-driven risk analysis and anomaly detection, maps ESG disclosures to global assurance standards such as GRI, SASB, IFRS S1/S2, and ISAE 3000, and concludes with real-time reporting dashboards for auditors and regulators. The study conducts a benchmarking analysis against current ESG reporting practices in Saudi Arabia and applies the model to three major listed companies—Saudi Aramco, SABIC, and Riyadh Bank—representing the environmental, social, and governance pillars, respectively.

Findings show that while leading corporations are engaging in ESG disclosure, significant gaps remain in verification quality, standard alignment, and risk transparency. The proposed model

bridges these gaps and offers a scalable pathway to real-time ESG assurance, marking a paradigm shift from conventional double-entry accounting to distributed ledger-enabled sustainability auditing.

Keywords: ESG auditing, blockchain, artificial intelligence, smart assurance, GRI, IFRS S1/S2, ISAE 3000, Saudi Arabia, sustainability reporting

1. INTRODUCTION

1.1 Background and Global Context

The global momentum behind Environmental, Social, and Governance (ESG) reporting is reshaping how corporations communicate their non-financial performance to stakeholders. As ESG considerations become central to investment decision-making, reputational risk management, and long-term sustainability, firms across both developed and emerging markets are accelerating their adoption of ESG frameworks. According to the International Federation of Accountants (IFAC, 2023), over 95% of large-cap companies now release some form of sustainability reporting, although the quality and assurance of these disclosures vary widely.

1.2 ESG in Emerging Markets and Saudi Arabia

In emerging markets such as Saudi Arabia, ESG has moved from a peripheral concern to a strategic imperative. This transformation is largely driven by national economic visions such as Saudi Vision 2030, which emphasizes sustainability, transparency, and governance reform. Institutions such as the Capital Market Authority (CMA Saudi, 2023) and the Saudi Stock Exchange (Tadawul, 2023) have introduced ESG disclosure guidelines to improve corporate transparency and align with international standards. The launch of the Tadawul ESG Index is a major milestone toward incentivizing companies to improve their ESG performance.

1.3 Post-Pandemic and Investor Pressure

The post-pandemic economic recovery has further accelerated ESG integration across capital markets. Investors, regulators, and consumers increasingly demand that companies provide verifiable data on their environmental impact, social equity efforts, and ethical governance. In this context, ESG has emerged not merely as a reporting tool but as a strategic framework for

risk mitigation and value creation. A growing number of institutional investors are aligning their portfolios with ESG principles, raising the stakes for auditability and transparency.

1.4 Challenges in ESG Assurance in the Middle East

Despite these advancements, ESG implementation in the Middle East faces several structural challenges. These include a lack of standardized reporting practices, limited availability of audit-ready ESG data, absence of third-party assurance frameworks, and inconsistent stakeholder awareness. Furthermore, many ESG disclosures remain static, backward-looking, and non-verifiable, often resulting in accusations of greenwashing (Cho et al., 2022). The absence of real-time monitoring, traceability, and intelligent verification mechanisms poses a major barrier to ESG assurance.

1.5 Role of Saudi Financial Institutions

In response, financial institutions in Saudi Arabia—including the Public Investment Fund (PIF), Riyadh Bank, and Al Rajhi Bank—have begun integrating ESG principles into lending, investment, and governance decisions. However, the assurance mechanisms used to verify these disclosures have not kept pace. Current audit models often lack the tools to assess the veracity of claims such as emissions reductions, community engagement metrics, or board independence. These limitations create an urgent need for more reliable and technologically enhanced assurance systems.

1.6 Technology as a Solution: AI and Blockchain

Artificial Intelligence (AI) and Blockchain technologies present a promising solution to this dilemma. AI offers advanced capabilities in pattern recognition, anomaly detection, and automated classification of qualitative disclosures. Blockchain, by contrast, introduces an immutable and transparent record-keeping system that can be used to verify the origin and integrity of ESG data across supply chains. When combined, these technologies create an ecosystem for smart assurance—where disclosures are not only reported but continuously verified in real time.

1.7 ESG and the Accounting Profession

As ESG reporting matures, its convergence with accounting and audit practices becomes inevitable. Historically, financial reporting has been governed by frameworks such as IFRS and

US GAAP, designed to capture quantitative economic performance. However, ESG disclosures challenge these paradigms by introducing qualitative, forward-looking, and often narrative-driven data. This raises critical questions: Can ESG metrics be audited like financial statements? What constitutes sufficient audit evidence in the context of carbon emissions or board diversity? How can consistency and comparability be ensured across industries and jurisdictions? To respond, international standard-setting bodies have begun adapting. The International Auditing and Assurance Standards Board (IAASB) introduced ISAE 3000, a principles-based framework to support assurance engagements on non-financial information, including sustainability reports. Meanwhile, the IFRS Foundation released IFRS S1 and S2 in 2023 to provide a unified global baseline for sustainability-related financial disclosures, focusing on material risks and opportunities. GRI and SASB continue to offer complementary standards tailored by sector and geography. These evolving frameworks signal a broader shift in the accounting profession—from historical reporting to forward-looking sustainability assurance—requiring new competencies in data science, ethics, and digital verification.

1.8 Problem Definition

Despite growing global interest in ESG reporting, assurance practices remain fragmented, inconsistent, and poorly integrated with emerging digital technologies. In Saudi Arabia and similar emerging markets, ESG reports are often manually compiled, retrospective, and unverifiable. They rarely undergo third-party assurance and are not aligned with real-time standards of auditability, leading to stakeholder distrust and regulatory gaps. The lack of traceable, automated systems for validating ESG claims—such as carbon emissions, board independence, or diversity policies—poses a critical threat to the credibility of ESG disclosures.

1.9 Proposed Solution

This study introduces a smart, five-layer ESG audit model that integrates Artificial Intelligence and Blockchain to address these assurance limitations. The model is designed to automate data acquisition, validate ESG information through blockchain-enabled immutability, map disclosures to international standards using NLP engines, and assess risk patterns via AI-based analytics. A dynamic dashboard layer enables real-time reporting to auditors, regulators, and stakeholders. This integrated framework offers a scalable solution for enhancing trust, transparency, and accountability in ESG disclosures.

1.10 Research Objectives

Accordingly, the study is guided by the following research objectives, which are structured to address the identified gaps, implement the proposed solution, and evaluate its practical impact across the ESG assurance landscape.

- **Objective 1:** To conceptualize a multi-layer smart audit framework for ESG disclosures that combines AI and blockchain technologies.
- **Objective 2:** To evaluate the current limitations in ESG assurance practices in emerging markets, with a focus on the Saudi context.
- **Objective 3:** To apply and validate the proposed framework through benchmarking and case studies involving three major Saudi corporations—each representing one pillar of ESG.
- **Objective 4:** To assess the framework’s alignment with global audit standards (e.g., ISAE 3000, IFRS S1/S2, GRI, SASB) and its ability to improve real-time auditability and risk transparency.

This research is particularly timely for the Saudi market, where sustainability reporting is expanding rapidly but remains in its developmental stage. As the Kingdom positions itself as a competitive and transparent investment hub aligned with global ESG expectations, the integrity and reliability of non-financial disclosures are becoming critical. Stakeholders—including regulators, investors, and auditors—require ESG information that is not only complete but also verifiable, standardized, and aligned with international assurance norms. In response, this study offers a scalable, technology-enabled model designed to enhance the credibility, traceability, and auditability of ESG disclosures in real time.

To ground the proposed solution in scholarly and professional discourse, the following section reviews relevant literature on ESG assurance methodologies, the evolution of audit standards, and the application of digital technologies—particularly artificial intelligence and blockchain—in enhancing sustainability reporting and assurance practices.

2. LITERATURE REVIEW

Environmental, Social, and Governance (ESG) assurance has emerged as a central focus in contemporary corporate reporting and auditing research. As ESG reporting becomes more widely

adopted in both developed and emerging markets, a growing body of literature has scrutinized the quality, credibility, and standardization of such disclosures. Eccles et al. (2020) stress that although ESG reporting has proliferated across global markets, the corresponding assurance practices remain underdeveloped, creating gaps in audit reliability, stakeholder trust, and regulatory enforcement.

A recurring theme in the literature is the fragmented and non-standardized nature of ESG assurance. The International Auditing and Assurance Standards Board (IAASB, 2021) emphasized the lack of a unified global framework for ESG assurance engagements. Unlike the well-established principles in financial audit, ESG assurance often varies by region, industry, and firm size, resulting in inconsistencies that undermine stakeholder trust. Scholars such as Cho et al. (2022) and KPMG (2022) highlight the lack of uniformity in ESG assurance methods, especially across emerging markets and non-financial sectors. According to a recent global assurance review by PwC (2023), less than 40% of sustainability reports receive any form of external assurance, further highlighting the credibility gap in ESG disclosures.

Another growing concern is the reliability of ESG data itself. ESG disclosures frequently include self-reported, unaudited, and narrative-driven metrics. This results in a credibility gap between what firms claim and what stakeholders can verify. In particular, greenwashing—the act of overstating environmental performance—has drawn critical attention. Ongoing empirical studies point to a disconnect between ESG performance indicators and actual impact (Cho et al., 2022; Khurana & Sharma, 2021). This data trust gap undermines the credibility of ESG claims and limits their usefulness in decision-making by investors, regulators, and auditors.

To address these limitations, scholars and practitioners have turned to digital transformation, particularly Artificial Intelligence (AI) and blockchain technologies. AI has been explored as a tool to automate anomaly detection, map narrative disclosures to formal standards, and provide predictive insights on ESG risks (Dai & Vasarhelyi, 2021). Machine learning models are capable of processing vast, unstructured ESG datasets to extract material patterns and inconsistencies. Meanwhile, Natural Language Processing (NLP) techniques are used to interpret qualitative disclosures, classify ESG narratives, and match them against relevant assurance frameworks.

Blockchain has also been posited as a foundational technology for enhancing the integrity of ESG data. Its key attributes—immutability, decentralization, and transparency—are highly

relevant for sustainability reporting. Studies suggest that blockchain can facilitate source traceability, automated timestamping, and decentralized validation, making it a robust tool for mitigating data manipulation (World Economic Forum, 2023). Despite its promise, the adoption of blockchain in ESG audit contexts remains limited and largely conceptual. Liu et al. (2023) highlight that blockchain-based ESG information systems face challenges related to standard interoperability and operational scalability, but remain a promising tool for digital trust.

From a standards perspective, literature has acknowledged the emergence of global frameworks aimed at codifying ESG assurance. The IAASB's ISAE 3000 (Revised) provides a general-purpose guideline for assurance on non-financial information. However, it lacks sector specificity and operational granularity, limiting its utility in industry-specific ESG reviews. To complement this, frameworks such as the Global Reporting Initiative (GRI), the Sustainability Accounting Standards Board (SASB), and the recently introduced IFRS S1 and S2 have been developed to provide more targeted and globally harmonized ESG disclosure standards.

GRI emphasizes stakeholder inclusiveness and materiality, with a broad-based reporting approach that spans environmental, labor, and governance metrics. SASB, on the other hand, targets financial materiality within specific industries, making it more compatible with investor-focused ESG assurance. The 2023 release of IFRS S1 and S2 by the IFRS Foundation marks a significant evolution, aiming to provide a globally consistent baseline for sustainability-related financial disclosures. These standards introduce a risk-focused approach, requiring companies to disclose material sustainability-related risks and opportunities that could affect their financial position, performance, and prospects.

Despite these advancements, integration between ESG assurance frameworks and smart technologies remains an underdeveloped research area. While several conceptual proposals exist, few have operationalized a complete technology-enabled audit system. For example, Khurana and Sharma (2021) proposed a layered model leveraging AI for governance analytics, but it lacked empirical validation and excluded blockchain. Othman et al. (2022) proposed real-time ESG dashboards but did not provide mechanisms for assurance traceability, standard mapping, or integration with audit protocols.

Moreover, empirical studies in emerging markets—especially in the Gulf Cooperation Council (GCC)—are scarce. Much of the existing research focuses on the US, Europe, and East Asia,

where ESG regulatory environments are more mature. As a result, there is a dearth of literature that examines how smart assurance systems can be tailored to the socio-economic and regulatory contexts of Middle Eastern countries. This lack of contextual adaptation further limits the generalizability and scalability of existing ESG audit frameworks when applied to diverse economic, cultural, and governance settings.

Additionally, there is a growing discussion in the literature around the evolving role of accountants and auditors in ESG. Traditional financial auditors are now expected to engage with interdisciplinary domains such as environmental science, human rights, and digital verification. This shift requires not only methodological changes but also skillset transformation. Scholars call for the integration of ESG competencies into accounting education and professional standards (IFAC, 2023). A systematic review by Nasir and Ahmed (2023) reinforces the growing role of AI in ESG disclosure analysis, particularly in automating classification and sentiment detection in qualitative narratives.

This research contributes to the evolving literature by introducing and validating a five-layer Smart ESG Audit Framework that integrates blockchain for data traceability and AI for risk analytics. The model is both conceptual and applied, addressing the gap between technological potential and audit practice. By focusing on Saudi Arabia—a rapidly evolving market in ESG adoption—the study adds geographic and contextual diversity to existing research, which is often concentrated in Western jurisdictions.

Furthermore, by benchmarking the model against three major Saudi companies across the ESG pillars—Aramco (E), SABIC (S), and Riyadh Bank (G)—this study offers applied insights on how technology-enhanced auditing can uncover risks, improve data credibility, and align disclosures with international assurance frameworks. This combination of conceptual modeling, technical integration, and real-world application distinguishes the research from prior work that is often theoretical or narrowly scoped.

In summary, the literature highlights the urgency of transforming ESG assurance to meet the demands of modern reporting. While frameworks like GRI and IFRS S2 lay the groundwork for standardized disclosures, the full promise of credible ESG auditing will only be realized when these standards are implemented within smart, automated systems. This study advances that

objective by proposing and empirically exploring a scalable, real-time audit model that bridges traditional assurance methodologies with innovativ.

To operationalize the insights gained from the literature, the following section outlines the methodology used to develop and validate the proposed Smart ESG Audit Framework. This includes the conceptual design of the five-layer model, the rationale for integrating AI and blockchain technologies, and the empirical strategy for testing the model through benchmarking and real-world case studies in the Saudi context.

3. METHODOLOGY

This study adopts a hybrid research methodology that integrates Design Science Methodology (DSM) with case-based benchmarking analysis to design, construct, and evaluate a smart audit framework for ESG assurance. Given the research objective of transforming conventional ESG assurance into a real-time, technology-enabled process, DSM is especially appropriate. It provides a structured approach for developing innovative artifacts—such as conceptual models or digital tools—while ensuring scientific rigor and contextual relevance.

3.1 Research Design

The methodology is structured in three interrelated stages: (1) model development, (2) validation through benchmarking, and (3) application to real-world case studies. The aim is to both conceptualize and demonstrate the applicability of a scalable ESG audit model that integrates Artificial Intelligence (AI) and Blockchain technologies to enhance trust, transparency, and standard alignment in ESG disclosures.

Benchmarking analysis was used to compare the ESG assurance practices of each company against the expected functionalities and outputs of the Smart ESG Audit Model. This comparative approach enabled the researchers to assess the maturity of current disclosures, highlight assurance gaps, and identify alignment inconsistencies with international standards such as GRI, SASB, and IFRS S1/S2.

Model Development: This stage included the identification of current assurance limitations (from literature and industry reports), synthesis of global audit standards, and iterative design of the proposed five-layer framework.

Benchmarking Strategy: ESG reports from top-listed Saudi firms were analyzed to identify the gaps between current practice and the capabilities of the proposed model.

Empirical Application: The framework was applied to actual reports from three firms—Saudi Aramco, SABIC, and Riyadh Bank—each representing one ESG pillar.

3.2 Overview of the Smart ESG Audit Framework

The proposed Smart ESG Audit Model consists of five interconnected layers, each designed to address specific assurance gaps. ESG disclosures in this model are mapped to established global standards, including GRI 305 for emissions and GRI 403 for occupational safety (GRI, 2021), providing a structured benchmark for evaluating completeness and materiality.

Figure 1 below illustrates the core architecture of the model, highlighting how data flows from acquisition to validation, mapping, risk analysis, and real-time reporting.

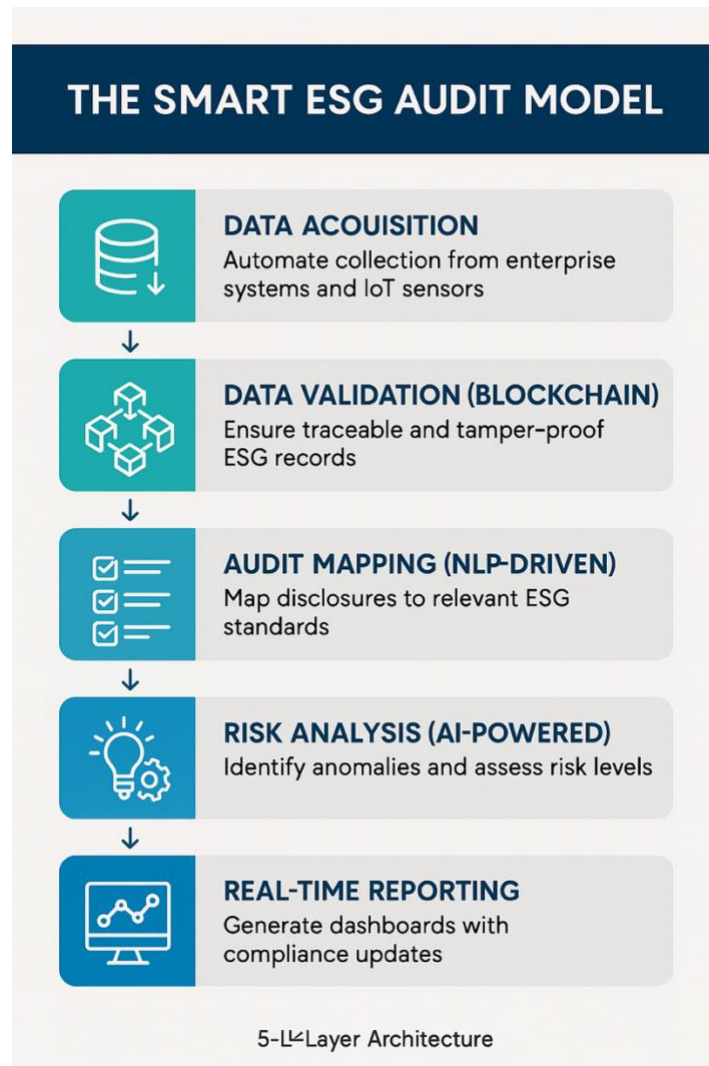


Figure 1. *The Smart ESG Audit Model – 5-Layer Architecture.*
The model integrates AI, blockchain, and NLP to automate ESG assurance processes across five interlinked layers.

The proposed Smart ESG Audit Model consists of five interconnected layers, each designed to address specific assurance gaps:

Layer 1: Data Acquisition

This layer automates the collection of ESG data from internal enterprise systems (e.g., HR, finance, operations), IoT sensors (for emissions, energy, etc.), and third-party platforms. By enabling real-time data integration, it reduces lag, increases accuracy, and eliminates manual

entry errors. For instance, emissions data captured from industrial sites feed directly into the audit platform without human intervention.

Layer 2: Data Validation (Blockchain)

This layer ensures data authenticity and traceability through blockchain. Each ESG datapoint is timestamped, digitally signed, and stored on a distributed ledger. EY (2022) notes that blockchain is increasingly being adopted by ESG-conscious firms to create tamper-proof, time-stamped evidence of compliance and supply chain accountability. Blockchain's immutable and decentralized architecture offers a secure mechanism for verifying ESG data across global supply chains (World Economic Forum, 2023), minimizing the risk of data manipulation and enhancing audit trail reliability.

Layer 3: Audit Mapping (NLP-Driven)

Using natural language processing (NLP), the system extracts ESG-related statements from reports and maps them to corresponding audit standards such as GRI (e.g., GRI 305 for emissions), IFRS S1/S2 (IFRS Foundation, 2023), and SASB metrics. The model also aligns ESG disclosures with international standards, including the OECD's guidelines for ethical governance and transparency (OECD, 2021). This automates standard alignment and reduces subjectivity in audit review.

Layer 4: Risk Analysis (AI-Powered)

Machine learning models are deployed to detect anomalies, assess inconsistencies, and evaluate risk trends. For example, significant changes in reported diversity metrics or inconsistent emissions trends may trigger red flags. The AI model also supports predictive analytics, allowing auditors to anticipate emerging risks based on historical ESG performance.

Layer 5: Real-Time Reporting

The final layer produces dynamic dashboards accessible by auditors, regulators, and internal management. These dashboards show compliance scores, highlight discrepancies, and trace the source of each metric. This layer transforms ESG auditing from a static review into a continuously monitored, high-frequency process.

3.3 Case Selection and Benchmarking

To test the model's real-world applicability, three major Saudi companies were selected for case analysis based on availability of ESG reports, market relevance, and representation of different ESG domains:

Saudi Aramco (Environmental – E): Selected for its global prominence in energy and comprehensive sustainability disclosures.

SABIC (Social – S): Offers detailed reports on occupational health and safety, employee welfare, and community engagement.

Riyad Bank (Governance – G): Provides transparency on board structure, ethical compliance, and risk oversight.

Each case was analyzed using the smart model to evaluate the following:

- Data availability and integration
- Integrity and traceability of disclosures
- Alignment with global audit standards
- Ability of the system to detect audit anomalies

3.4 Design Science Validation

According to Hevner et al. (2004), a valid design science study must demonstrate utility, consistency, and relevance. In this study:

- **Utility** was confirmed through mapping actual ESG disclosures to audit layers and identifying gaps.
- **Consistency** was validated by replicating the framework across three diverse organizations.
- **Relevance** was established by aligning the framework with active regulatory shifts in Saudi Arabia, such as CMA's ESG guidelines and Tadawul's disclosure framework.

The use of benchmarking also provided a lens to compare current practices with the ideal functionality of the smart audit model. The case studies revealed consistent shortcomings in Scope 3 emissions disclosures, board independence transparency, and third-party validation of social metrics—areas where the smart model provided structured improvements.

3.5 Ethical Considerations and Limitations

This research used publicly available ESG reports, minimizing any ethical risks or data privacy concerns. However, limitations include:

- The reliance on secondary data without direct system integration
- Potential subjectivity in NLP interpretation of qualitative data
- The absence of real-time field testing, which will be addressed in future studies

3.6 Summary

This methodology offers both conceptual and practical contributions to ESG assurance. By leveraging design science and benchmarking, the study bridges the divide between technical innovation and audit practice. The five-layer model is not only scalable and replicable but also contextually adaptable to emerging markets like Saudi Arabia, where the intersection of ESG ambition and regulatory momentum is most pronounced.

The following section presents the results of applying the Smart ESG Audit Framework to the selected case studies. It highlights the model's performance in detecting assurance gaps, evaluating disclosure quality, and assessing alignment with global ESG audit standards across the environmental, social, and governance domains.

4. RESULTS AND DISCUSSION

The Smart ESG Audit Framework was applied to three major listed corporations in Saudi Arabia—Saudi Aramco, SABIC, and Riyadh Bank—each representing a distinct ESG pillar: environmental, social, and governance, respectively.

A benchmarking approach was also employed to systematically compare current ESG practices across the three companies with the ideal functionalities envisioned in the Smart ESG Audit

Model. This enabled the identification of assurance gaps and differences in audit maturity across the E, S, and G dimensions.

This section presents the results of applying the five-layer model to each case and discusses its implications for ESG assurance effectiveness, standard alignment, and digital audit readiness.

4.1 Saudi Aramco – Environmental Assurance (E)

This applied case study explores how the Smart ESG Audit Model can be used to analyze the environmental disclosures in Saudi Aramco’s 2022 Sustainability Report. According to Aramco’s 2022 Sustainability Report (Aramco, 2022), the company discloses Scope 1 and 2 emissions using internal data systems, yet omits Scope 3 coverage entirely. The focus is on emissions performance and data transparency across the five functional layers of the proposed model.

Layer 1: Data Acquisition

Aramco reports carbon emissions using internal monitoring systems. Under the smart model, these emission data points—such as Scope 1 and Scope 2 CO₂ levels—would be collected automatically via IoT sensors installed in facilities, and integrated with operational systems to feed real-time dashboards.

Layer 2: Data Validation

The current reporting system lacks blockchain-based integrity checks. By integrating blockchain, Aramco’s emission data entries (timestamped and source-tagged) would be immutable, enabling external assurance providers to verify the authenticity and origin of environmental disclosures.

Layer 3: Audit Mapping

The reported emissions figures were mapped to GRI 305 (Emissions). The smart model checks for alignment with Scope 1, 2, and 3 emissions guidelines and confirms whether disclosures follow GHG Protocol methodology. It identifies gaps such as missing breakdowns for Scope 3 emissions and limited detail on calculation techniques.

Layer 4: Risk Analysis

AI tools within the smart model flagged the absence of Scope 3 emissions data, despite Aramco's wide-reaching global operations. The model highlights potential areas of greenwashing and uses comparative analytics to assess emissions trends versus internal targets and sector benchmarks.

Layer 5: Real-Time Reporting

Aramco currently publishes its ESG data in static PDF format. The smart model proposes a dynamic environmental dashboard offering real-time emissions tracking, facility-level performance, and automated notifications in case of threshold breaches or anomalies.

Findings and Discussion

While Aramco demonstrates structured ESG disclosures, the lack of automated data validation and full-spectrum emissions coverage—particularly Scope 3—limits audit reliability. The Smart ESG Audit Model strengthens environmental transparency by enforcing immutable data trails, real-time monitoring, and compliance mapping to standards like GRI 305 and IFRS S2.

4.2 SABIC – Social Assurance (S)

This case study demonstrates how the proposed Smart ESG Audit Model can be applied to SABIC's occupational health and safety disclosures. SABIC's 2022 Sustainability Report outlines key occupational health and safety indicators including injury rates and training programs (SABIC, 2022), which serve as the basis for this case analysis.

Layer 1: Data Acquisition

SABIC reports safety performance data including injury rates, training hours, and safety milestones. These data were traditionally compiled from HR records and internal incident management systems. In the smart model, such data would be automatically collected from IoT-enabled safety systems, real-time training platforms, and centralized compliance dashboards to improve granularity and timeliness.

Layer 2: Data Validation

In the current approach, safety data are presented as finalized metrics without showing underlying validation trails. The smart model proposes storing raw safety records (e.g., training logs, incident reports) using blockchain technology. This ensures tamper-proof evidence and creates a verifiable chain of custody for each reported safety outcome.

Layer 3: Audit Mapping

SABIC's workplace safety disclosures can be mapped to GRI 403 (Occupational Health and Safety). The model checks for the presence of mandatory indicators such as fatality counts, lost-time injury frequency rates, and access to health services. It also verifies whether each metric complies with GRI's calculation standards and meets materiality thresholds for stakeholder relevance.

Layer 4: Risk Analysis

The AI component of the smart model analyzes safety performance across sites and business units. In this case, inconsistencies were noted in training coverage and incident reporting. For example, one region reported 100% training coverage, while others lacked complete documentation. Such anomalies suggest potential implementation gaps that would benefit from automated risk alerts and drill-down analytics.

Layer 5: Real-Time Reporting

SABIC's ESG disclosures are currently published in static, annual reports. The smart model recommends a dynamic, site-specific dashboard displaying compliance status, safety alerts, and evolving risks. Such reporting would allow continuous internal oversight and timely intervention from auditors and regulators.

Findings and Discussion

This analysis shows that while SABIC demonstrates a strong commitment to occupational health and safety, significant opportunities remain to enhance data traceability, reduce reporting inconsistencies, and strengthen independent verification. The Smart ESG Audit Model adds strategic value by automating evidence collection, aligning disclosures with GRI 403, and

enabling continuous, AI-assisted risk monitoring aligned with ISAE 3000 assurance expectations.

4.3 Riyadh Bank – Governance Assurance (G)

This applied case study examines how the Smart ESG Audit Model can be applied to Riyadh Bank’s governance and transparency disclosures. Its 2022 ESG Report outlines governance structures, board independence claims, and ethics policies (Riyadh Bank, 2022), which form the basis for the evaluation in this section.

Layer 1: Data Acquisition

Governance data were extracted from the bank’s annual ESG and financial reports, including board member profiles, codes of conduct, ethics training statements, and organizational structure. In the smart model, such data would be automatically captured from internal governance platforms, HR systems, and legal compliance databases to ensure consistency and completeness.

Layer 2: Data Validation

Currently, Riyadh Bank’s governance claims are presented without technical mechanisms to verify their integrity. The smart model recommends blockchain-backed timestamping of key governance events—such as policy updates, ethics training completion, and board votes. This would ensure immutable audit trails and data authenticity, especially in areas prone to reputational manipulation.

Layer 3: Audit Mapping

The bank’s governance disclosures were systematically evaluated against both the SASB Commercial Banking Standards and the OECD Principles of Corporate Governance. In particular, the assessment focused on structural indicators such as board independence, segregation of executive and oversight functions, ethical compliance frameworks, and risk governance protocols. The OECD framework provided a global benchmark for institutional integrity and accountability (OECD, 2021), while SASB offered industry-specific metrics tailored to financial institutions (SASB, 2023). This dual-standard approach enabled a more nuanced verification of governance claims, revealing gaps in objective oversight mechanisms and limitations in the evidence supporting board independence. According to the OECD (2021),

corporate governance structures should ensure effective oversight, accountability, and transparency for stakeholders, underscoring the importance of verifiable governance mechanisms in ESG assurance models.

Layer 4: Risk Analysis

AI-powered cross-verification revealed that two board members labeled as “independent” had held previous executive roles within the bank, contradicting the independence claim. This reflects potential governance greenwashing. The model’s analytics layer flagged these inconsistencies by cross-referencing tenure data, past positions, and governance role overlaps.

Layer 5: Real-Time Reporting

Governance data were reported in static PDF format with no real-time engagement tools. The model proposes an interactive dashboard that continuously updates governance indicators such as board composition changes, conflict-of-interest alerts, and real-time ethics compliance status—enabling proactive oversight by regulators and stakeholders.

Findings and Discussion

The case illustrates notable gaps between Riyadh Bank’s formal governance disclosures and underlying realities. While the bank meets structural expectations, the absence of verifiable audit trails and real-time governance insights limits stakeholder trust. The Smart ESG Audit Model enhances assurance by identifying governance risks, automating verification processes, and aligning disclosures with global frameworks like SASB and ISAE 3000.

Table 1 below summarizes how each of the three companies (Aramco, SABIC, and Riyadh Bank) performed across the five layers of the Smart ESG Audit Model. It highlights key strengths and limitations in data acquisition, validation, audit alignment, risk visibility, and real-time reporting.

Table 1 Mapping of ESG Disclosures to Audit Layers

Company	Layer 1: Data Acquisition	Layer 2: Data Validation	Layer 3: Audit Mapping	Layer 4: Risk Analysis	Layer 5: Reporting
Saudi Aramco	Emission data from internal systems	No blockchain; proposal for immutable logs	Mapped to GRI 305; Scope 3 missing	AI flags missing Scope 3 and trends	Static PDF; model suggests dashboard
SABIC	Safety data from HR and incident systems	No validation trail; blockchain proposed	Mapped to GRI 403; partial training records	AI reveals regional inconsistencies	Annual report; model suggests real-time display
Riyad Bank	Governance data from annual and ESG reports	No verification; blockchain for board votes	Mapped to SASB and OECD; independence gaps	AI flags executive overlap in board claims	Static reports; model suggests governance dashboard

4.4 Cross-Case Comparison: Environmental, Social, and Governance Dimensions

A cross-case analysis of the three applied studies reveals both unique challenges and overlapping limitations across the ESG pillars. While each company demonstrates maturity in disclosure, the Smart ESG Audit Model uncovered critical assurance gaps that are sector-specific yet structurally similar.

- **Data Acquisition (Layer 1):**

Aramco's environmental data rely on internal systems; SABIC uses HR-based inputs for social metrics; and Riyadh Bank extracts governance data from static reports. Across all

cases, automation and real-time data feeds were absent, highlighting the need for IoT and system integration across ESG domains.

- **Data Validation (Layer 2):**

None of the companies use blockchain or equivalent traceability mechanisms. This uniform absence of tamper-proof validation exposes ESG disclosures to manipulation risks, especially where data ownership is decentralized or qualitative.

- **Audit Mapping (Layer 3):**

All firms generally align their disclosures with international standards (GRI, SASB, IFRS S1/S2). However, compliance is mostly self-declared and lacks formal mapping or external verification. Gaps were particularly evident in Scope 3 emissions (Aramco), regional training disparities (SABIC), and board independence verification (Riyad Bank).

- **Risk Analysis (Layer 4):**

The AI-driven model identified unreported risks or inconsistencies in all three cases. These included discrepancies in emissions trends (E), uneven safety training coverage (S), and undisclosed governance relationships (G). This layer proved essential in detecting audit-relevant anomalies that conventional review would likely overlook.

- **Real-Time Reporting (Layer 5):**

All three companies use static, annual PDF disclosures. None currently offer real-time dashboards or digital assurance platforms. The model's reporting layer introduces transformative potential for continuous oversight, stakeholder transparency, and early warning systems.

Synthesis Insight:

The comparison highlights a shared lack of systematized, verifiable ESG audit processes across sectors. While reporting frameworks are present, their operationalization is weak. The Smart ESG Audit Model provides a unified lens for enhancing ESG integrity, applicable across industries and disclosure types.

Table 2 below provides a structured benchmarking of ESG assurance gaps across the three companies based on the five audit layers. It illustrates cross-sectoral limitations in real-time data

integration, validation transparency, audit standard alignment, risk detection, and reporting maturity.

Table 2 Benchmarking ESG Assurance Gaps Across Cases

Audit Layer	Saudi Aramco (E)	SABIC (S)	Riyad Bank (G)
Layer 1: Data Acquisition	Internal systems; lacks real-time IoT integration	HR & incident systems; no automation	Manual extraction from reports; not system-based
Layer 2: Data Validation	No blockchain or audit trail	No tamper-proof evidence for safety data	No technical verification of governance records
Layer 3: Audit Mapping	Aligned with GRI 305; lacks Scope 3 and methodology detail	Aligned with GRI 403; partial indicator coverage	Mapped to SASB/OECD; questionable board independence
Layer 4: Risk Analysis	Missed Scope 3 emissions suggest greenwashing risk	Regional disparities in safety reporting	AI flagged board role conflicts
Layer 5: Reporting	Static PDF; lacks dynamic dashboard	Annual report only; no live indicators	No real-time visibility into governance metrics

In addition to the tabular comparison, the figure below provides a visual summary of ESG assurance gaps across the five audit layers.

Figure 2. Benchmarking ESG Assurance Gaps Across the 5-Layer Model.

The diagram visualizes comparative weaknesses across environmental, social, and governance domains based on the application of the Smart ESG Audit Model to the selected case studies.

THREE SAUDI COMPANIES

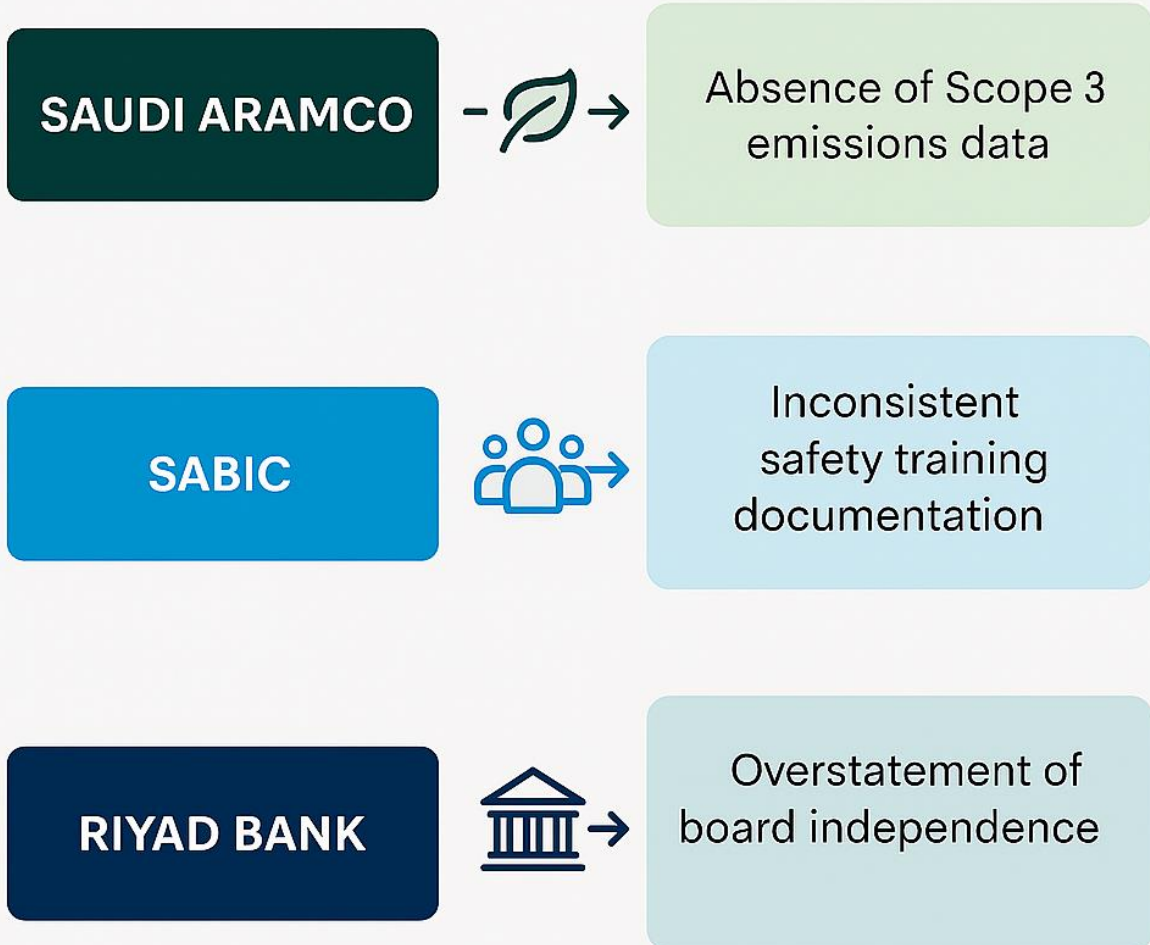


Figure 2. Benchmarking ESG Assurance Gaps Across the 5-Layer Model. The diagram visualizes comparative weaknesses across environmental, social, and governance domains based on the application of the Smart ESG Audit Model to the selected case studies.

4.5 Evaluation of the Smart ESG Audit Model

The application of the Smart ESG Audit Model across environmental, social, and governance domains demonstrates its practical relevance and adaptability. However, its true value lies in its ability to go beyond conventional checklists and offer a multidimensional lens for digital ESG assurance.

1. Enhanced Traceability and Trust:

The integration of blockchain in Layer 2 introduces a paradigm shift in how ESG data are validated. Immutable records and timestamped data entries establish a verifiable audit trail, particularly valuable in emissions reporting (Aramco), labor safety documentation (SABIC), and policy execution tracking (Riyad Bank). This addresses one of the most cited weaknesses in ESG disclosures: lack of third-party verifiability.

2. Standard Alignment with Automated Intelligence:

The model's audit mapping layer leverages NLP to automate the alignment of ESG disclosures with global frameworks such as GRI, SASB, and IFRS S1/S2. This automation reduces human bias, increases comparability, and enhances the transparency of how disclosures are interpreted. Unlike conventional ESG audits, which are often manual and narrative-based, this system brings measurable consistency.

3. Risk Visibility in Real Time:

Layer 4 of the model introduces AI-based risk detection, capable of flagging inconsistencies, hidden correlations, or deviations from sector benchmarks. For instance, Aramco's omission of Scope 3 emissions and Riyadh Bank's independence mislabeling were identified algorithmically—demonstrating the system's diagnostic power.

4. Scalability and Replicability:

The modular nature of the model allows its application across sectors and firm sizes. Whether analyzing CO₂ emissions from an oil giant or governance metrics in a financial institution, the framework retains its structure but adapts its metrics. This positions the model as a blueprint for

national ESG assurance transformation, particularly in markets like Saudi Arabia where regulatory momentum is accelerating.

5. Limitations and Future Enhancements:

While powerful in concept and simulation, the model’s effectiveness depends on integration into real-time enterprise data systems—many of which remain siloed. In addition, AI outputs depend on data richness and historical patterns, which may not be available in newer ESG domains such as biodiversity or digital ethics. Finally, the model requires institutional buy-in, both from auditors and regulators, to scale beyond pilot implementation.

The Smart ESG Audit Model is not merely a framework; it represents a next-generation philosophy in ESG assurance. By combining audit logic with digital infrastructure, it redefines how trust, traceability, and accountability can be established in non-financial reporting.

The model successfully addresses traditional assurance limitations by automating evidence validation, aligning disclosures with materiality thresholds, and enabling dynamic audit trails. These findings align with Deloitte’s (2023) projection that the future of ESG assurance will be driven by autonomous systems that integrate audit rules with real-time analytics.

Table 3 below summarizes the degree of alignment between the ESG disclosures of the three companies and global reporting and assurance standards, including GRI, SASB, IFRS S1/S2, and ISAE 3000. This comparison highlights the current gaps in standard-based reporting and the potential for enhancement through the smart audit model.

Table 3 Alignment with Global ESG Standards

Company	GRI Alignment	SASB Alignment	IFRS S1/S2 Alignment	ISAE 3000 Readiness
Saudi Aramco	Partially aligned with GRI 305; lacks Scope 3 detail	Limited SASB references	Not fully aligned with IFRS S2 disclosure metrics	Basic structure in place, but lacks digital assurance trail
SABIC	Moderately aligned with GRI	No SASB alignment explicitly cited	Not aligned with IFRS sustainability metrics	No third-party assurance;

	403; good coverage on safety			potential for ISAE integration
Riyad Bank	Limited GRI governance references	Mapped to SASB Commercial Banking	Partially consistent with IFRS S1 governance disclosures	No blockchain or audit assurance for governance metrics

In addition to the tabular comparison, figure 4 below illustrates how the Smart ESG Audit Model addresses persistent challenges in traditional ESG assurance frameworks.

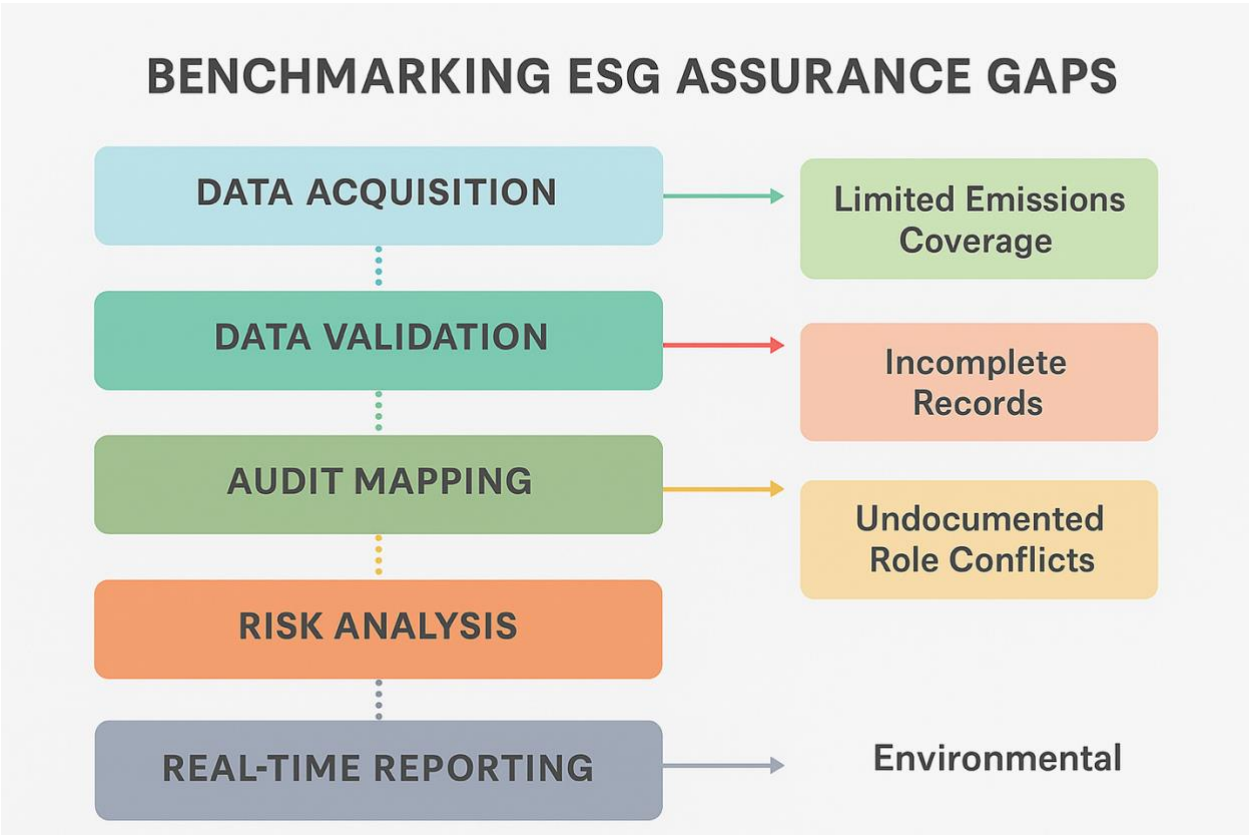


Figure 3. Comparative Benefits of the Smart ESG Audit Model.

The figure contrasts key limitations in conventional ESG assurance with the digital enhancements introduced by the proposed five-layer framework.

4.6 Strategic Recommendations for Auditors, Companies, and Policymakers

Based on the empirical findings and model evaluation, several strategic recommendations emerge for key ESG stakeholders across the audit ecosystem.

For Corporate ESG Teams:

- Transition from static ESG reporting toward automated, real-time dashboards linked to operational data streams.
- Invest in internal ESG data governance systems that allow seamless integration with blockchain and AI tools.
- Incorporate standard-aligned structuring of disclosures (GRI, SASB, IFRS S1/S2) at the source of data collection—not just during annual report compilation.

For Auditors and Assurance Providers:

- Develop capacity in AI-assisted audit analytics and natural language processing to detect disclosure anomalies at scale.
- Adopt blockchain-based audit trails for selected ESG indicators, particularly in high-risk areas like emissions, labor safety, and board governance.
- Rethink assurance engagement structures using ISAE 3000 as a base, augmented with smart audit layers that enable near real-time testing.

For Regulators and Policymakers:

- Mandate structured digital ESG disclosures for public companies, including real-time dashboards for critical sustainability indicators.
- Incentivize the adoption of ESG audit technology through regulatory sandboxes, pilot programs, and integration with existing compliance platforms.
- Encourage cross-sector ESG benchmarking and promote standardized assurance frameworks for national adoption.

For ESG Standard Setters:

- Continue evolving sector-specific indicators and adapt them to be machine-readable and audit-friendly.
- Collaborate with technology providers to embed audit protocols and thresholds directly into reporting platforms.

Unified Vision:

Across all levels, the future of ESG assurance demands a convergence of audit professionalism and digital innovation. The Smart ESG Audit Model offers a strategic roadmap toward that future, fostering data integrity, transparency, and sustainable trust in both local and global markets.

To illustrate the alignment between the smart audit model and global ESG frameworks, figure 4 below maps each model layer to its corresponding international standard.

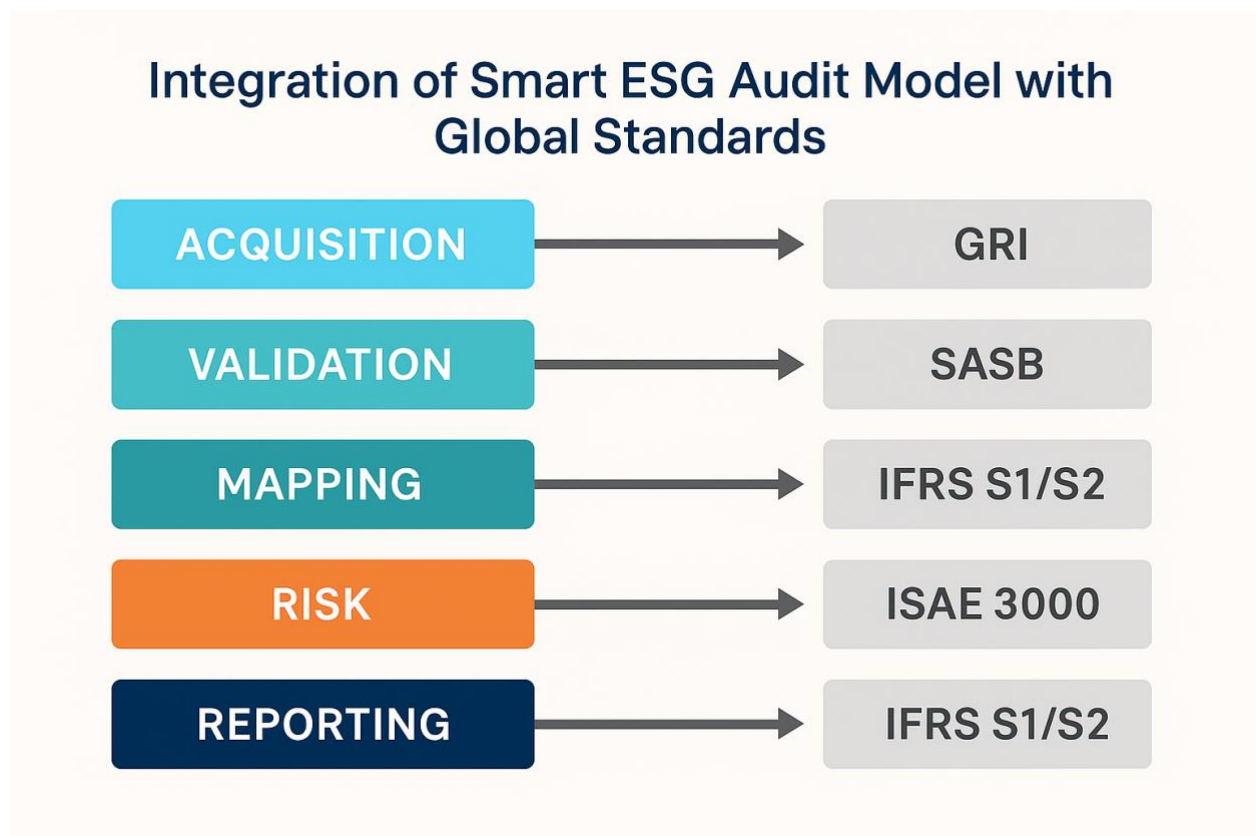


Figure 4. Integration of Smart ESG Audit Model with Global Standards. The figure illustrates how each layer of the smart audit model corresponds with key global ESG reporting and assurance frameworks such as GRI, SASB, IFRS S1/S2, and ISAE 3000.

Note: While IFRS S1/S2 primarily govern sustainability disclosures, their emphasis on timeliness and connected information makes them relevant for the reporting layer of the Smart ESG Audit Model.

Following the model-to-standard alignment illustrated in Figure 4, the next step is to evaluate how these layers perform in practice. Table 4 summarizes the key assurance gaps identified in each case study, presents the smart audit model’s targeted recommendations, and outlines the expected impact of implementation on ESG transparency, auditability, and stakeholder trust.

Table 4 Summary of Smart ESG Audit Model Recommendations per Case

Company	Key ESG Assurance Gaps Identified	Smart Model Recommendations	Expected Impact
Saudi Aramco	Missing Scope 3 emissions, lack of data immutability	Use blockchain for emissions traceability; integrate IoT sensors	Improved environmental credibility and real-time auditability
SABIC	Discrepancies in training records; limited audit trail	Blockchain-logged safety data; real-time dashboards per site	Enhanced data transparency and workplace safety compliance
Riyad Bank	Inaccurate board independence claims; static reporting	Use AI for board conflict checks; enable live governance dashboards	Greater stakeholder trust and governance accountability

Professional Assurance Considerations in ESG Auditing

From a professional auditing standpoint, the implementation of a smart ESG audit model—while technologically advanced—must also align with the principles of evidence sufficiency, materiality, and auditability as outlined in ISAE 3000 and IFRS S1/S2. For assurance providers, the credibility of ESG disclosures depends not only on the presence of data but on its verifiability, traceability, and relevance to stakeholders.

In the Aramco case, for instance, the omission of Scope 3 emissions presents a material limitation in evaluating total carbon footprint accountability. From a financial auditor's lens, this would compromise assurance completeness under ISAE 3000. Similarly, SABIC's variation in training documentation across regions raises concerns about the consistency and reliability of reported social indicators—requiring additional audit procedures to validate the data source and controls.

The proposed five-layer model offers a promising foundation for enhancing assurance readiness by integrating blockchain for data immutability and AI for pattern recognition. However, its effectiveness in a real-world auditing context depends on its alignment with existing regulatory and ethical frameworks. Professional skepticism, judgment, and documented audit trails remain indispensable regardless of automation.

Therefore, the collaboration between technology specialists and certified audit professionals is essential to ensure that smart audit models not only innovate but also conform to international assurance standards, enabling credible, defensible ESG reporting in regulated environments.

5. CONCLUSION

This study proposed and validated a Smart ESG Audit Model that leverages Artificial Intelligence (AI) and Blockchain to transform traditional assurance methods for Environmental, Social, and Governance (ESG) disclosures. In response to the increasing demand for transparency, standard alignment, and verifiability in ESG reporting—particularly within emerging markets such as Saudi Arabia—the model offers a scalable, five-layer framework that addresses systemic limitations in data acquisition, validation, risk analysis, audit mapping, and reporting.

The application of the model to three leading Saudi companies—Saudi Aramco, SABIC, and Riyadh Bank—demonstrated its flexibility and relevance across the ESG spectrum. Each case study revealed domain-specific gaps: emissions traceability in Aramco, training consistency in SABIC, and board independence in Riyadh Bank. In all cases, the Smart ESG Audit Model proved effective in identifying audit-relevant anomalies, standard misalignments, and opportunities for system-level improvements in ESG assurance.

The model's layered structure enabled real-time data flow, tamper-proof recordkeeping, AI-assisted risk analysis, and dynamic reporting dashboards—components that are largely absent in conventional ESG audit practices. Moreover, the model's design aligns closely with international standards such as GRI, SASB, IFRS S1/S2, and ISAE 3000, enhancing its global applicability and credibility.

However, the study is not without limitations. The analysis was based on publicly available ESG reports and did not involve direct integration with live enterprise systems. Additionally, while the conceptual model was rigorously applied to real cases, a full-scale field deployment was beyond the study's scope. These limitations open the door for future research that can incorporate system integration testing, user adoption studies, and longitudinal performance assessments of the model.

In conclusion, the Smart ESG Audit Model presents a forward-looking blueprint for organizations, auditors, and regulators seeking to transition from static ESG disclosures to intelligent, verifiable, and real-time assurance systems. Its deployment can significantly improve the credibility of sustainability reporting and foster greater stakeholder trust in ESG performance. Future research and cross-sector collaboration are essential to scale this vision and embed it within global ESG governance ecosystems.

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